Beyond Single-Exponent Scaling: Multi-Parameter Entropy–Information Relations in Nonlinear Field Systems

Christof Krieg

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Abstract

Previous work established an empirical entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}$$

for the Critical Coherence Index (CCI) in one-, two-, and three-dimensional nonlinear field systems [2, 3, 4]. While this single-exponent description works well in low dimensions, recent three-dimensional results revealed a breakdown of a unique effective exponent and the emergence of an exponent plateau.

In this work we introduce a multi-parameter scaling ansatz,

$$\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{a} I_{\mathrm{nn}}^{-b},$$

and test it across dimensions. We find that the relative contributions of entropy production and structural information vary systematically with dimension, indicating that the entropy–information balance is governed not by a universal single exponent but by a dimension-dependent scaling family. In particular, the entropy contribution changes sign across dimensions, revealing a qualitative shift in the role of dynamical activity.

These results suggest that higher-dimensional systems require a multi-channel structural description, consistent with the broader interpretation of the entropy–information relation as a projection of a higher-dimensional structural free-energy landscape. All results are empirical; a first-principles derivation remains open.

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1 Introduction

Recent work established that the Critical Coherence Index (CCI) is consistent with an empirical entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}.$$

In one and two dimensions, this relation is well described by a single effective exponent [2, 3]. In three dimensions, however, the notion of a unique exponent breaks down and is replaced by a broad plateau of near-equivalent exponents [4].

This suggests that the single-exponent ansatz is only an effective low-dimensional approximation. The present paper tests a more general multi-parameter scaling model.

2 Scaling Ansatz

We consider the generalised relation

$$\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{a}\, I_{\mathrm{nn}}^{-b}, \tag{1}$$

which implies the log-linear form

$$\ln \mathcal{C}_{\mathrm{CCI}} = c_0 + a \ln \dot{S}_{\mathrm{cg}}^{(+)} + b(-\ln I_{\mathrm{nn}}). \tag{2}$$

The single-exponent ansatz $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ is formally recovered when $a = 1$ and $b = \alpha$. The goal is to determine whether the two-parameter description improves the effective scaling description across dimensions.

3 Method

For each dimension we use the corresponding ensemble data:

- 1D: `cci_entropy_information_test.csv`
- 2D: `2d_cci_entropy_information_test.csv`
- 3D: `3d_cci_entropy_information_test.csv`

We fit Eq. (2) by linear regression in log-space and extract the coefficient a for entropy production, the coefficient b for structural information, and the coefficient of determination R^2.

4 Results

4.1 Multi-Parameter Scaling Fit

We fit the generalised scaling relation

$$\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{a}\, I_{\mathrm{nn}}^{-b}$$

using a log-linear regression. The resulting coefficients are:

$$\text{1D:} \quad a \approx -1.6, \quad b \approx 0.33, \quad R^2 \approx 0.76 \tag{3}$$

$$\text{2D:} \quad a \approx -0.08, \quad b \approx 0.50, \quad R^2 \approx 0.74 \tag{4}$$

$$\text{3D:} \quad a \approx 0.30, \quad b \approx 0.30, \quad R^2 \approx 0.37 \tag{5}$$

4.2 Dimensional Restructuring of Scaling Weights

Figure 1 shows the evolution of the coefficients across dimensions.

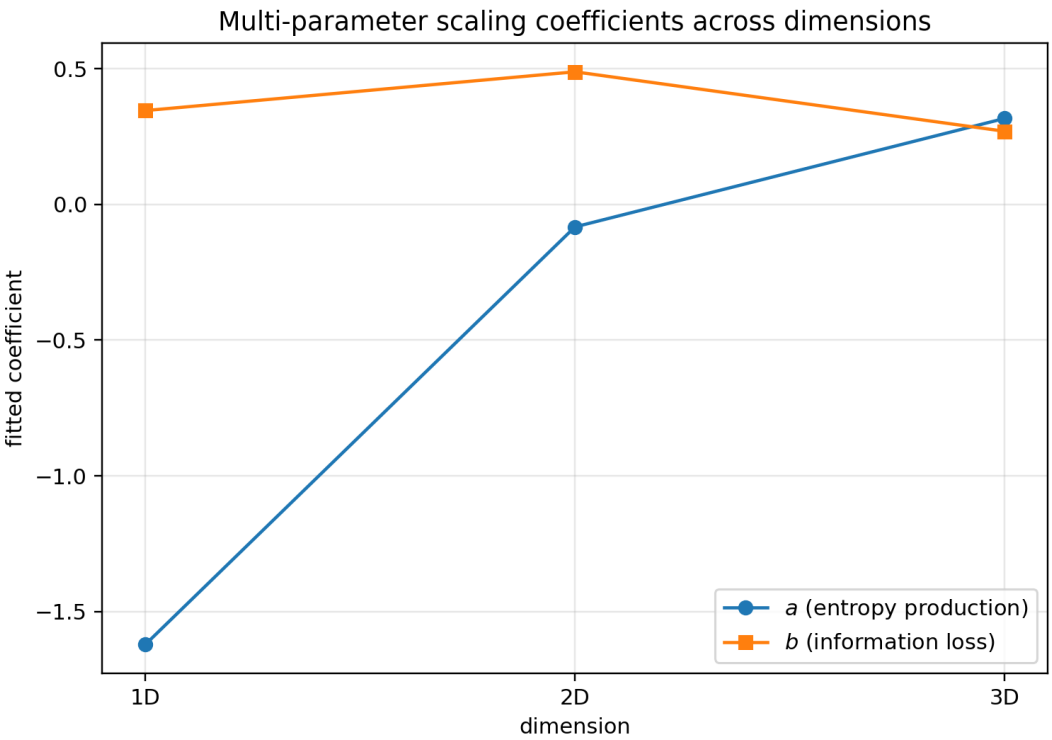

Figure 1: Multi-parameter scaling coefficients across dimensions. The entropy-production weight a changes sign from negative in 1D to positive in 3D, while the information-loss weight b peaks in 2D. This indicates a qualitative restructuring of the scaling relation.

A clear dimensional trend is observed:

- In **1D**, entropy production enters with a strong negative weight ($a < 0$), indicating that increased activity is associated with reduced CCI.
- In **2D**, the entropy contribution becomes negligible, while structural information dominates (b maximal).
- In **3D**, both contributions become comparable, indicating a balanced multi-channel regime.

4.3 Breakdown of Predictive Power

The quality of the fit decreases significantly in 3D. Table 1 summarises the coefficients and R^2 values.

Table 1: Multi-parameter scaling coefficients across dimensions.

Dimension	a	b	R^2
1D	-1.60	0.33	0.76
2D	-0.08	0.50	0.74
3D	$+0.30$	0.30	0.37

This is confirmed by the observed vs. predicted comparison (Fig. 3).

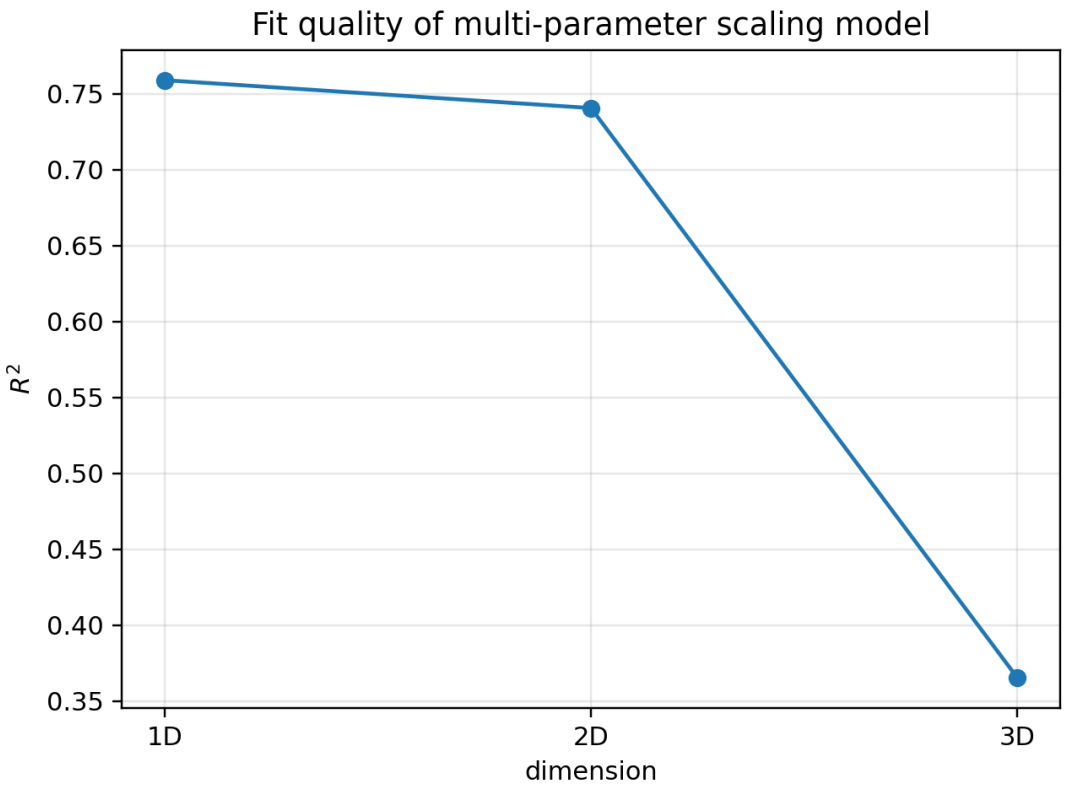

Figure 2: Coefficient of determination R^2 across dimensions. While the multi-parameter model provides a strong description in 1D and 2D, its predictive power is significantly reduced in 3D.

4.4 Interpretation

The results reveal a qualitative restructuring of the scaling relation:

- The entropy contribution changes sign across dimensions.
- The dominance of structural information peaks in 2D.
- In 3D, the system transitions to a balanced but weakly predictive regime.

This suggests that the entropy–information scaling is governed by a dimension-dependent family of relations rather than a single universal form. The exponent plateau observed in 3D is naturally explained as a projection artifact of an underlying multi-parameter scaling structure.

5 Discussion

The multi-parameter ansatz provides a natural interpretation of the 3D exponent plateau observed previously [4]. Rather than a failure of the entropy–information scaling, the plateau signals that the effective instability relation can no longer be compressed into a single exponent.

In low dimensions, the system is sufficiently constrained that a one-parameter scaling description is adequate. In higher dimensions, additional channels of structural reorganisation and decorrelation become relevant, leading to a family of effective scaling relations.

The transition from negative to positive entropy weight ($a < 0$ in 1D to $a > 0$ in 3D) indicates a qualitative change in the role of dynamical activity: from a destabilising effect in low dimensions to a contributing factor in higher-dimensional coherence dynamics. This sign flip is

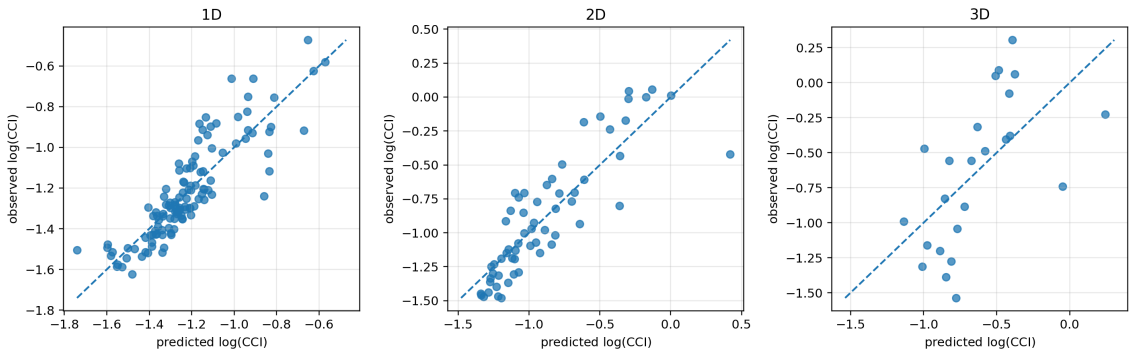

Figure 3: Observed vs. predicted $\ln(\mathcal{C}_{\mathrm{CCI}})$ for each dimension. The agreement is strong in 1D and 2D, but significantly weaker in 3D, indicating that even the two-parameter scaling model is insufficient to fully capture the higher-dimensional structure.

not a numerical artefact but reflects a genuine restructuring of the instability mechanism across dimensions.

Within the broader framework [1], this supports the interpretation that the entropy–information relation is a projection of a higher-dimensional structural free-energy landscape. The single-exponent law is then a low-dimensional reduction of a more general multi-parameter structure.

6 Conclusion

We have extended the entropy–information scaling analysis of the CCI beyond the single-exponent ansatz by introducing the multi-parameter relation

$$\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{a}\, I_{\mathrm{nn}}^{-b}.$$

The results indicate that the relative weights a and b vary systematically with dimension. This suggests that nonlinear field systems are governed not by a universal exponent, but by a dimension-dependent scaling family.

Future work will investigate whether this family can be derived from the structural free energy or from a more general multi-channel instability principle.

All results are empirical. A first-principles derivation remains an open problem.

A Reproducibility

All code and data are available at:

https://github.com/Chris4081/structural-selection-principle

Step 1 — Install dependencies:

```
pip install numpy pandas matplotlib scipy scikit-learn
```

Step 2 — Generate the required CSV data files (see Papers 10 and 11 for full details):

```
python cci_alpha_scaling.py        # → cci_entropy_information_test.csv
python cci_entropy_scaling_2d.py   # → 2d_cci_entropy_information_test.csv
python cci_entropy_scaling_3d.py   # → 3d_cci_entropy_information_test.csv
```

Step 3 — Run the multi-parameter fit:

```
python multiparameter_fit.py
```

Output files:

- `paper13_ab_coefficients.png` — Figure 1: a and b vs dimension
- `paper13_r2_comparison.png` — Figure 2: R^2 across dimensions
- `paper13_observed_vs_predicted.png` — Figure 3: observed vs predicted per dimension

References

[1] C. Krieg, *A Structural Free Energy on the Solution Manifold*, preprint, available from the author (2026).

[2] C. Krieg, *Entropy–Information Scaling of the Critical Coherence Index in Nonlinear Field Dynamics*, preprint, available from the author (2026).

[3] C. Krieg, *Entropy–Information Scaling of the Critical Coherence Index in Two-Dimensional ϕ^4 Field Dynamics*, preprint, available from the author (2026).

[4] C. Krieg, *Persistence of Entropy–Information Scaling in Three Dimensions*, preprint, available from the author (2026).

[5] C. Krieg, *Structural Selection — companion code*, https://github.com/Chris4081/structural-selection-principle (2026).